SURESH KUMAR N

Generative AI Engineer · LLM Applications & Agentic AI

Python · LangGraph · LangChain · RAG · LLM Evaluation (RAGAS) · Fine-Tuning (LoRA) · AWS Bedrock · Full Stack

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PROFESSIONAL SUMMARY

Generative AI Engineer with 3.5 years of experience designing and shipping production LLM systems on secure, bilingual (English/Arabic) e-government platforms for the Government of Oman, serving 18,900+ suppliers across 117 countries. Builds **LangGraph multi-step agents** with tool calling, **RAG pipelines** with hybrid search and reranking, and a **trust layer** of LLM evaluation (RAGAS, LLM-as-judge), hallucination guardrails and tracing (LangSmith/Langfuse). Experienced with open-source model fine-tuning (LoRA) and serving (Ollama, vLLM) alongside hosted models (AWS Bedrock, OpenAI GPT-4o, Anthropic Claude, Google Gemini), chosen per task on accuracy, latency and cost. Ships AI features end to end with Python/FastAPI, Node.js/GraphQL and Angular/React. Impact: **67% faster contract review, 50% faster search, 40% less manual review effort**.

CORE GENAI COMPETENCIES

Agentic AI	LangGraph (stateful multi-step agents, self-check & retry loops), LangChain, AI Agents, Tool/Function Calling
RAG & Retrieval	RAG Pipelines, Embeddings, Chunking Strategies, Hybrid Search (vector + keyword), Rerankers, MongoDB Atlas Vector Search, ChromaDB, Citations & Source Grounding
LLM Evaluation & Safety	RAGAS, LLM-as-Judge, Groundedness & Relevance Scoring, Offline Eval Suites (retrieval hit rate), Hallucination Guardrails, LangSmith, Langfuse
LLMs & Model Ops	OpenAI GPT-4o, Anthropic Claude, Google Gemini, Hugging Face, Open-Source LLMs, LoRA Fine-Tuning, Ollama, vLLM, AWS Bedrock, Model Selection
LLM Engineering	Prompt Engineering, Document Summarisation & Extraction, Document Classification, Token & Cost Optimisation (context trimming, prompt compaction)

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript (ES6+), SQL, Java, HTML5, SCSS
Backend & APIs	FastAPI, Node.js, Express.js, GraphQL, REST APIs, Microservices, Spring Boot, Redis
Frontend	React.js, Redux, Angular (v9–v22, Signals, SSR, RxJS), Tailwind CSS
Databases	MongoDB (Mongoose), Vector Databases (Atlas Vector Search, ChromaDB), MySQL, Schema Design, Indexing
Cloud, DevOps & Testing	AWS (EC2, S3, Bedrock), Docker, Jenkins CI/CD, Git, Playwright (AI-assisted E2E), Vitest, Jasmine, Postman
Practices	JWT, RBAC, i18n & Arabic RTL, WCAG, SEO, Agile (Scrum), Jira

PROFESSIONAL EXPERIENCE

Software Engineer (Generative AI & Full Stack) — Business Gateways International, Chennai **Jun 2023 – Present**

Software Engineer (Dec 2023 – Present); Software Engineer Intern (Jun 2023 – Dec 2023). Bringing LLM, RAG and agentic AI capabilities into core workflows of bilingual (English/Arabic) e-government platforms for the Government of Oman.

RABT — AI-Enabled Industrial Supplier Registration & Certification Platform **Jan 2026 – Present**

Python, LangGraph, LangChain, RAG, Hybrid Search, Rerankers, MongoDB Atlas Vector Search, RAGAS, LangSmith/Langfuse, Node.js, GraphQL, Angular 22 SSR, MySQL, Playwright

- Designed **LangGraph agentic workflows** with tool calling that automate certificate validation, certificate updates and product-approval checks, **reducing manual review effort by 40%**.
- Built a **RAG semantic search pipeline** (LangChain + embeddings + MongoDB Atlas Vector Search) over 20K+ supplier, product and certificate records, **cutting search time by 50%**.
- Raised retrieval quality with tuned chunking strategies, **hybrid search (vector + keyword)** and a **reranker**; reduced LLM spend through token and cost optimisation (context trimming, prompt compaction).
- Built a Python **LLM evaluation pipeline** with RAGAS and LLM-as-judge scoring, added **hallucination guardrails** on agent outputs, and set up end-to-end tracing with LangSmith/Langfuse.
- Exposed AI and platform features through Node.js/Express REST and GraphQL APIs on MySQL for 432 industrial organisations at p95 latency under 300 ms.

- Led migration of the public website from Next.js to Angular 22 (SSR, Signals; 12 routes, 47 components) in 1 week with 59 Vitest unit tests and **AI-assisted Playwright E2E tests**; Lighthouse performance 65 → 90+.

JSRS — Joint Supplier Registration System **Jun 2023 – Dec 2025**

Python, LLM APIs, Embeddings, Hugging Face, LoRA, Ollama, vLLM, AWS Bedrock, Spring Boot, Angular, React, Redux, MySQL, Redis, Docker, AWS

- Built an **LLM-powered contract review service** that summarises contracts and extracts key dates, values and obligations, cutting review time from 30 to 10 minutes per contract (**67% faster**).
- Developed **LLM- and embedding-based document classification** that auto-routes 500+ supplier documents/month at **90% accuracy**, replacing manual sorting.
- Evaluated and integrated **open-source Hugging Face models** in Python (local inference with Ollama/vLLM, **LoRA fine-tuning**) alongside hosted models on **AWS Bedrock**, selecting models per task on accuracy, latency and cost.
- Delivered 5 Spring Boot microservice modules (Enterprise Admin, Company Profile, Contract Management, JStore, Sky Cards) for 18,981 suppliers and 135,044 products/services across 117 countries, secured with Spring Security, JWT and RBAC.
- Optimised MySQL with indexing (dashboard load 6 s → 2 s) and added Redis caching (API latency −35%); built Angular and React + Redux UIs with English/Arabic RTL; deployed with Docker, AWS EC2/S3 and Jenkins CI/CD at 75% unit-test coverage.

Full Stack Engineer (Part-time) — NoviTech R&D Pvt Ltd, Chennai (Remote) **May 2022 – Jun 2023**

- Built end-to-end web applications with Angular, Node.js/Express REST APIs and MySQL schema and query design.

GENAI PROJECTS

DataOps AI Assistant — Agentic RAG over Technical Documentation **github.com/sureshnarayananankumar/dataops-ai-assistant**

Python 3.11, FastAPI, LangGraph, ChromaDB, React, Google Gemini / OpenAI

- Built a question-answering assistant over dbt, Snowflake, SQL and pipeline-runbook documents (PDF, TXT, Markdown, SQL) with an ingestion pipeline: text extraction → chunking → embeddings → ChromaDB.
- Designed a **LangGraph agent** (query understanding → retrieval → generation → evaluation → response) with **one automatic retry** using an improved query and more context when an answer is not supported by the sources.
- Added **LLM-as-judge** relevance and groundedness checks on every answer, **numbered citations** (document, PDF page, chunk, match score) and an honest "I don't know" response when documents lack the answer.
- Built an **offline evaluation suite** (test dataset + report on retrieval hit rate and groundedness), a FastAPI backend, a React dashboard, and a configurable LLM provider (Gemini by default, OpenAI optional).

Portfolio AI Copilot — In-Browser RAG Agent & JD Fit Analyser **sureshnarayananankumar.github.io/portfolio**

- Built an in-browser retrieval agent that answers recruiter questions from resume content with citations, plus a job-description analyser that scores skill fit with evidence; vanilla JavaScript, no frameworks.

Full Stack Projects **github.com/sureshnarayananankumar**

- **HR ATS** (React, NestJS, Prisma, MySQL, Redis): 12-stage hiring pipeline, 6 roles / 70 permissions, MFA, audit logs. **AssessHub** (Java 21, Spring Boot, React 18): timed assessments with auto-grading, 55 JUnit + 19 Vitest tests.

CERTIFICATIONS

Generative AI — NASSCOM and MeitY **2026**

AI Full Stack Masterclass — NoviTech R&D Private Limited, Chennai **2023**

EDUCATION

B.Sc. Information Technology — Yadava College (Madurai Kamaraj University), Madurai (83%) **2019 – 2022**